

New Branch Rollout

Terhubung ke server soal via nc 43.153.208.219 3160. Diberikan alokasi network 192.168.50.0/24 yang harus dibagi menggunakan VLSM berdasarkan kebutuhan host, dialokasikan dari kebutuhan terbesar ke terkecil:

- Dept A: 100 host
- Dept B: 50 host
- Dept C: 20 host
- Link WAN (point-to-point): 2 host

Perhitungan VLSM dilakukan manual sebagai berikut:

1. Dept A (100 host):

Alokasi subnet mask: /25 (kapasitas 126 usable host, block size 128)

Network Address: 192.168.50.0/25 (Range IP: .0 – .127)

2. Dept B (50 host):

Alokasi subnet mask: /26 (kapasitas 62 usable host, block size 64)

Network Address: 192.168.50.128/26 (Range IP: .128 – .191)

3. Dept C (20 host):

Alokasi subnet mask: /27 (kapasitas 30 usable host, block size 32)

Network Address: 192.168.50.192/27 (Range IP: .192 – .223)

4. Link WAN point-to-point (2 host):

Alokasi subnet mask: /30 (kapasitas 2 usable host, block size 4)

Network Address: 192.168.50.224/30 (Range IP: .224 – .227)

Sehingga untuk jawaban pertanyaan 1 adalah 192.168.50.128/26, sedangkan untuk pertanyaan 2 adalah 192.168.50.224/30.

```
(sofi@sofi)-[~]
$ nc 43.153.208.219 3160

=====
IP Allocation Request - Cabang Baru
=====

Kantor cabang baru dapat alokasi network 192.168.50.0/24.
Kebutuhan host per segmen (VLSM, alokasikan dari kebutuhan
terbesar ke terkecil, mulai dari awal blok):
- Dept A : 100 host
- Dept B : 50 host
- Dept C : 20 host
- Link WAN point-to-point (router-to-router) : 2 host

Jawab pake format CIDR network address, contoh: 192.168.50.0/25

Q1) Berapa network address subnet untuk Dept B (50 host)? > 192.168.50.128/26
Benar.

Q2) Berapa network address subnet untuk link WAN (2 host)? > 192.168.50.224/30
Benar.

Semua jawaban benar. Flag: FLAG{vlsn_math_checks_out}
```

Setelah kedua jawaban dikirim ke server dan dinyatakan benar, ditampilkan flag yaitu FLAG{vlsn_math_checks_out}.

New Office Setup

Terhubung ke simulasi switch (SW-CORE01) via nc 43.153.208.219 3161. Task: buat 2 VLAN untuk memisahkan traffic Sales & IT, dengan trunk port untuk uplink. Konfigurasi dilakukan langsung di CLI simulasi menggunakan syntax Cisco IOS standar:

```
(sofi@sofi)-[~]
$ nc 43.153.208.219 3161
SW-CORE01 (simulated switch)

Task: kantor baru butuh 2 VLAN buat misahin traffic Sales & IT,
plus uplink trunk ke switch lain.
- VLAN 10 name SALES, port Fa0/1 access ke VLAN 10
- VLAN 20 name IT, port Fa0/2 access ke VLAN 20
- Fa0/24 jadi trunk port (uplink)

Verifikasi konfigurasi lo pake 'show vlan brief' atau
'show running-config' setelah selesai.

Switch> ena
Switch# conf t
Switch(config)# vlan 10
Switch(config-vlan)# name SALES
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name IT
Switch(config-vlan)# exit
Switch(config)# int fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
Switch(config)# int fa0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
Switch(config)# int fa0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
Switch(config)# end
```

Verifikasi dilakukan dengan show vlan brief, menampilkan VLAN 10 (SALES) di fa0/1 dan VLAN 20 (IT) di fa0/2 sudah aktif sesuai spec. Sistem otomatis memvalidasi konfigurasi sebagai benar dan menampilkan flag yaitu FLAG{vlan_trunk_deployed_correctly}.

```
Switch# show vlan brief
VLAN Name                Status    Ports
--
1    default              active
10   SALES                 active    fa0/1
20   IT                    active    fa0/2

! Configuration verified: compliant with deployment spec.
FLAG{vlan_trunk_deployed_correctly}
Switch#
```

Blind Spot

Login SSH ke server (netadmin:CorpNet2026!) di port 3162. Skenario: ada dashboard monitoring yang hanya bisa diakses dari localhost server itu sendiri, namun perlu diakses dari luar.

```
(sofi@sofi)-[~]
$ ssh netadmin@43.153.208.219 -p 3162
The authenticity of host '[43.153.208.219]:3162 ([43.153.208.219]:3162)' can't be established.
ED25519 key fingerprint is: SHA256:id7z8uKC3bKHv5Ey+y38wXc1vmH7pN80t5lBYHD99Ac
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[43.153.208.219]:3162' (ED25519) to the list of known hosts.
netadmin@43.153.208.219's password:
Welcome to Alpine!

The Alpine Wiki contains a large amount of how-to guides and general
information about administrating Alpine systems.
See <https://wiki.alpinelinux.org/>.

You can setup the system with the command: setup-alpine

You may change this message by editing /etc/motd.

7c4ef87f5c6d:~$ ss -tulnp
Netid    State      Recv-Q     Send-Q     Local Address:Port      Peer Address:Port      Process
udp      UNCONN     0           0           127.0.0.11:48133        0.0.0.0:*
tcp      LISTEN     0           5           127.0.0.1:8080          0.0.0.0:*
tcp      LISTEN     0          128           0.0.0.0:22             0.0.0.0:*
tcp      LISTEN     0         4096           127.0.0.11:43205        0.0.0.0:*
tcp      LISTEN     0          128           [::]:22                [::]:*

7c4ef87f5c6d:~$ curl http://127.0.0.1:8080
-bash: curl: command not found
7c4ef87f5c6d:~$ exit
logout
Connection to 43.153.208.219 closed.
```

Investigasi awal menggunakan ss -tulnp untuk melihat service yang listen, ditemukan port 8080 listening di 127.0.0.1 saja (bukan 0.0.0.0), artinya service tersebut memang sengaja dibatasi hanya bisa diakses secara lokal. Percobaan curl langsung di dalam server gagal karena tool tersebut tidak tersedia.

Solusinya adalah "meminjam" akses tersebut menggunakan SSH local port forwarding, dengan membuka koneksi baru:

```
(sofi@sofi)-[~]
$ ssh -L 8888:127.0.0.1:8080 netadmin@43.153.208.219 -p 3162
netadmin@43.153.208.219's password:
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```

Perintah ini meneruskan port 8888 di komputer lokal ke port 8080 di server (yang hanya bisa diakses dari localhost server). Setelah tunnel terbentuk, dashboard berhasil diakses dari luar dengan curl `http://localhost:8888`, menampilkan halaman "Internal Ops Dashboard" beserta flag-nya yaitu `FLAG{tunnel_vision_localhost_bound}`.

```
(sofi@sofi)-[~]
$ curl http://localhost:8888
<!DOCTYPE html>
<html>
<head><title>Internal Ops Dashboard</title></head>
<body>
<h1>Internal Ops Dashboard</h1>
<p>This panel is intentionally bound to 127.0.0.1 only and is not reachable f
rom outside this host.</p>
<p>Flag: FLAG{tunnel_vision_localhost_bound}</p>
</body>
</html>
```

Chain of Trust

Login SSH ke bastion host (junior:Welcome123!) di port 3163. Skenario: server core-db diisolasi total dari internet dan hanya bisa diakses lewat satu bastion host tertentu.

```
(sofi@sofi)-[~]
$ ssh junior@43.153.208.219 -p 3163
The authenticity of host '[43.153.208.219]:3163 ([43.153.208.219]:3163)' can't be established.
ED25519 key fingerprint is: SHA256:7axq8AaWnkuTrLkMs90Yy8ex0QjUqPspqre0y1TaLl4
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[43.153.208.219]:3163' (ED25519) to the list of known hosts.
junior@43.153.208.219's password:
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```

Investigasi dimulai dengan mengecek `~/.ssh/config`, `known_hosts`, dan `/etc/hosts` di bastion untuk mencari petunjuk target internal, namun tidak ditemukan informasi langsung.

```

0b26b921a01e:~$ ls -la ~/.ssh
total 16
drwx--S--- 2 junior junior 4096 Aug 13 02:30 .
drwxr-sr-x 1 junior junior 4096 Aug 18 09:12 ..
-rw----- 1 junior junior 1410 Aug 14 14:12 known_hosts
-rw-r--r-- 1 junior junior 97 Aug 13 02:29 known_hosts.old
0b26b921a01e:~$ cat ~/.ssh/config 2>/dev/null
0b26b921a01e:~$ cat ~/.ssh/known_hosts 2>/dev/null
internal-target ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHx0UjFw7Ql3c13zDeeeaDCap3nV2HtgYKojC0u6MGbJ
internal-target ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDQQL46yLdmPWInBRDkhkursBQ55PIy+AUaVA1YtgUjGhCm2X6boLE1S6lhURiw2
PLE5f5B3vxnKX5I/UGYG0Q++o9GDF6liM9kqWmMYZf6T1xL6sdEXxRiSez/nJ5yHxz/80L/gNeNoLhgDG0rCa3ZAqBJY1582BXs8sg0wZZP10ymPT8Fn
cV7VNL/L0NUSFingpb4Gt70lbX00aESfCrDHC7HfA3xz5wiq37shQ7u134n2XgCcgPqmrP5is7hI0vA4s33owTQknxpQDUVRkK3qaiDBWD3KQD6A7ey
lpyGp75iguYjd5ULlrYc1cp5nEbRg+DyPDxu9n1Ikq/r3g/aDognwvDbiAIfME04zgG5tbro9QvULjk0kV60Z3dXec402PZ6EjTfaD+2s1KU+IRccbk
fa/mTW7NVipdEbFtGH9J1JbBBvA3Q68ahc3Y9pAVKq9rB4Yr8ud4u4yrh460Iirf66kkRBLEDIXuynag4wxLUo2RF0pUmIRouLdvKNM=
internal-target ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBGh17XN6B8B1IlqbedrewHb4GmIWN
gkCCqaB234kPqfLgXmhbRdb0JCEg8z08xiEXQDrrY0+3HkkziCS36o9Rhxo=
172.20.0.2 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHx0UjFw7Ql3c13zDeeeaDCap3nV2HtgYKojC0u6MGbJ
43.153.208.219 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIKfmB+Ix0gfNsm4KMUL5CePw/HtCaIb0DR857o0EY+E5
[43.153.208.219]:3163 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPLRJayI+Bs9Ef6a9GU/Y+dL4R85G0K0yygEslnrGC6d
172.20.0.3 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPLRJayI+Bs9Ef6a9GU/Y+dL4R85G0K0yygEslnrGC6d
172.21.0.5 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPLRJayI+Bs9Ef6a9GU/Y+dL4R85G0K0yygEslnrGC6d
172.21.0.1 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIKfmB+Ix0gfNsm4KMUL5CePw/HtCaIb0DR857o0EY+E5
0b26b921a01e:~$ cat /etc/hosts
127.0.0.1 localhost
::1 localhost ip6-loopback
fe00:: ip6-localnet
ff00:: ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
172.20.0.3 0b26b921a01e
172.21.0.5 0b26b921a01e
0b26b921a01e:~$ ls -la ~
total 32
drwxr-sr-x 1 junior junior 4096 Aug 18 09:12 .
drwxr-sr-x 1 root root 4096 Aug 9 03:19 ..
-rw----- 1 junior junior 9602 Aug 18 09:12 .bash_history
drwx--S--- 2 junior junior 4096 Aug 13 02:30 .ssh
-rw-r--r-- 1 junior junior 275 Aug 8 19:19 note.txt

```

Ditemukan file note.txt di home directory yang berisi catatan dari Senior Network Engineer, menjelaskan bahwa server internal-target hanya bisa dijangkau dari bastion ini, lengkap dengan kredensial (dbadmin:DbSecure!2026) dan saran untuk menggunakan SSH ProxyJump / -J.

```

0b26b921a01e:~$ cat note.txt
Note from Senior Network Engineer:

The core-db server has no direct route from the internet, it can only be
reached from this bastion host.

Host: internal-target
User: dbadmin
Pass: DbSecure!2026

Use this bastion as a jump host to reach it (look into SSH ProxyJump / -J).

```

Karena bastion memiliki akses langsung ke jaringan internal, cukup dilakukan SSH biasa dari dalam bastion menuju target:

```
0b26b921a01e:~$ ssh dbadmin@internal-target
dbadmin@internal-target's password:
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03fa46bdc293:~$ ls -la
total 24
drwxr-sr-x  1 dbadmin  dbadmin    4096 Aug 13 11:45 .
drwxr-xr-x  1 root    root       4096 Aug  9 03:19 ..
-rw-----  1 dbadmin  dbadmin    1848 Aug 18 09:22 .bash_history
drwx--S---  2 dbadmin  dbadmin    4096 Aug 17 02:19 .ssh
-rw-----  1 dbadmin  dbadmin     31 Aug  9 03:52 flag.txt
03fa46bdc293:~$ cat flag.txt
FLAG{hop_through_the_bastion}}
```

Setelah masuk, `ls -la` menampilkan `flag.txt` di home directory `dbadmin`, yang kemudian dibaca dengan `cat flag.txt`. Ditemukan flag `FLAG{hop_through_the_bastion}`.